



AR CAMPUS ASSISTANT

Campus navigation from outdoor routes to indoor facilities, schedules, shuttle context, and future AR guidance.

Problem

NUS students often reach the correct building but still struggle to find classrooms, lifts, restrooms, printers, study spaces, and shuttle connections.

MILESTONE 1 SCOPE

- 20 user stories covering students, visitors, accessibility, transport, safety, and map updates.
- 10 ideated product and engineering features.
- Full-stack prototype: React frontend, Go backend, SQLite data, JWT auth, deployment.

Core Features

- 1 **Auth**
email, demo, Google Sign-In, JWT

3 **Search**
place lookup and route tracing

5 **Facilities**
buildings, rooms, lifts, printers

7 **NUSMods**
sync status and trigger

9 **Mobile**
responsive web and iOS shell
- 2 **Map**
MapLibre campus rendering

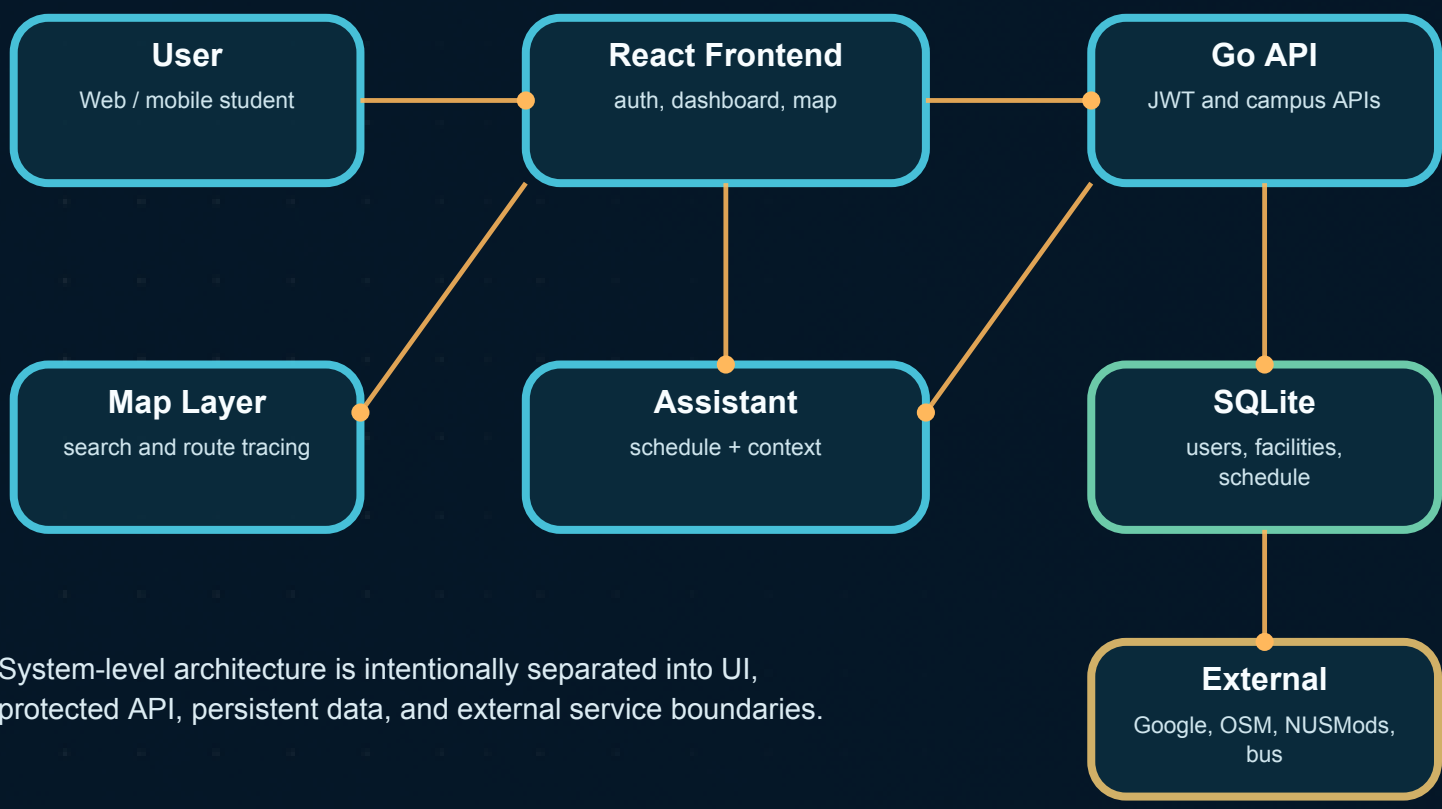
4 **Assistant**
schedule-aware recommendations

6 **Schedule**
view, create, delete items

8 **Bus**
stops, arrivals, active vehicles

10 **SWE**
issues, branches, CI/CD, tests

Architecture



Standard Diagrams

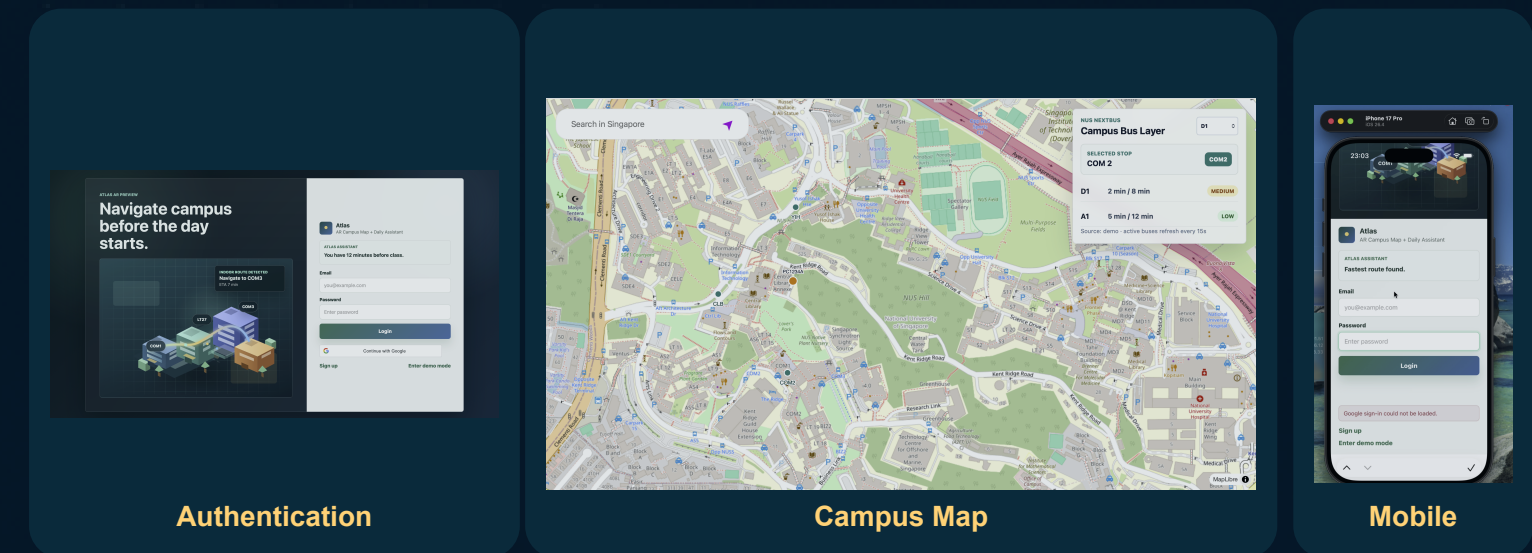
- 1 **Sequence**
Login -> backend verification -> JWT -> protected dashboard.

2 **Activity**
Search location -> select place -> center map -> draw route.

3 **ER**
users, buildings, facilities, schedule items, sync status.

4 **Architecture**
frontend, backend, database, external campus services.

Prototype Screens



The demo flow shows login, protected dashboard access, schedule context, facility filtering, responsive mobile layout, and campus map interaction.

Testing and CI/CD

- Backend** cd backend && go test ./...

Frontend cd frontend && npm run build

iOS shell npm run ios:sync

Deploy docker compose up -d --build

PIPELINE GATES
GitHub Actions plan: checkout, npm ci, frontend build, Go tests, Docker build, then deployment validation on the hosted Render app.

User Story Coverage

- Navigation**

AR map, indoor guidance, travel time, saved locations.
- Campus Life**

study spaces, events, printing, less crowded facilities.
- Access**

wheelchair-friendly routes, lifts, accessible entrances.
- Transport**

nearby shuttle stops, routes, arrivals, bus context.
- Safety**

emergency exits, first-aid points, security offices.
- Community**

student corrections and administrator review.

Walkthrough and Links

- 1 Open deployed app and login through demo/auth flow.

2 Show protected dashboard, schedule, recommendations, and facility filters.

3 Show map rendering, search-oriented interaction, and route tracing.

4 Explain repo logs, testing commands, CI/CD plan, and deployment setup.

App: orbital-artemis-armap-nus.onrender.com
GitHub: [77chenchen/Orbital_Artemis_ArMap_Nus](https://github.com/77chenchen/Orbital_Artemis_ArMap_Nus)